



WELDING PROCEDURE SPECIFICATION (WPS)

Company Name **RENEWCOILS ENGINEERING AND SUPPLY CO. LTD**

WPS No. **RNC-WPS-015**

Date **29-Mar-13**

Revision No. **0**

Supporting PQR No.(s) **RNC-PQR-015**

Date **29-Mar-13**

Type(s) **Manual**

Welding Process(es) **Gas Tungsten Arc Welding + Shield Metal Arc welding**
(GTAW + SMAW)

JOINT (QW-402)

Joint Design **Single V**

Backing (Yes) ☐ (No) ☒ X

Backing Material (Type) **N/A**

☐ Metal ☐ Nonfusing Metal
☐ Nonmetallic ☐ Other

Notes _____

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. **4** Group No. **1** to P-No. **4** Group No. **1**

Material Specification Type and Grade **A335 P11, F182-F11, A234 WP11** TO **A335 P11, F182-F11, A234 WP11**

Chem. Analysis and Mech. Prop. _____

Thickness Range :

Base Metal : Groove **1.6-16.36 mm.** Fillet **ALL**

Pipe Dia. Range : Groove **ALL** Fillet **ALL**

Other _____

FILLER METALS (QW-404)

	GTAW	SMAW
Spec. No. (SFA)	5.28	5.5
AWS No. (Class)	ER80S-G	E8016-B2
F-No.	6	4
A-No.	3	3
Size of Filler Metals	Ø 2.4 ~ 3.2mm.	Ø 2.6 ~ 4.0 mm.
Deposited Weld Metals	3 mm.	5.18 mm.
Thickness Range :		
Groove	6 mm.	10.36 mm.
Fillet	ALL	ALL
Electrode-Flux (Class)	-	-
Flux Trade Name	-	-
Consumable Insert	-	-
Other	Kobelco - TGS-1CM	Kobelco - CMA 96

POSITION (QW-405)

Position(s) of Groove **ALL**

Welding Progression Up ☒ X Down ☐

Position(s) of Fillet **ALL**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **704°C~746°C**

Time Range **Holding Time 2 Hour.**

Notes **Heating Rate=220°C Max./Cooling Rate=250°C Max**

PREHEAT (QW-406)

Preheat Temp. Min. **200°C~250°C**

Interpass Temp. Max. **300°C**

Preheat Maintenance **N/A**

Notes **Resistance heater**

GAS (QW-408)

Percent Composition

	Gas(es)	(Mixture)	Flow Rate
Shielding	Argon	9.99%	12-16 Ltr/Min
Trailing	-	-	-
Backing	Argon	-	-



WELDING PROCEDURE SPECIFICATION (WPS)

WPS No.

RNC-WPS-015

Rev.

0

ELECTRICAL CHARACTERISTICS (QW-409)

Current ☒ AC ☒ DC Polarity EN(GTAW)+EP(SMAW)
Amps (Range) 120-150 (GTAW) + 100-140 (SMAW) Volts (Range) 9-12(GTAW) + 10-17(SMAW)
Tungsten Electrode Size and Type Ø 2.4 mm, Thoriated (EWTh-2)
(Pure Tungsten, 2% Thoriated, etc.)
Mode of Metal Transfer for GMAW N/A
(Spray arc, short circuiting arc, etc.)
Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

String or Weave Bead BOTH APPLICATIONS
Orifice or Gas Cup Size 8.0 mm - 19.0 mm
Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING
Method of Back Gouging N/A
Oscillation N/A
Contact Tube to Work Distance N/A
Multiple or Single Pass (per side) MULTIPASS
Multiple or Single Electrodes SINGLE ELECTRODE
Travel Speed (Range) 4-17 Cm/Minute
Peening N/A
Other N/A

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1st	GTAW	ER80S-G	2.4 mm.	DC EN	120-150	8-12	4-6	
2nd	GTAW	ER80S-G	2.4 mm.	DC EN	120-150	10-13	8-1	
3 & Over	SMAW	E8016-B6	3.2 mm.	AC/DCEP	100-140	18-28	10-17	

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of ASME-IX, 2010 latest Revision.

Prepared by :

Mr. Chatchawan

Name

Signed

11/04/13

Date

Reviewed by :

ภูชิน สีหอมไชย

Phuchin Seehomchai

Name

Signed

11/04/13

Date

Approved by :

ภูชิน สีหอมไชย

Phuchin Seehomchai

Name

Signed

11/04/13

Date



Procedure Qualification Record (PQR)

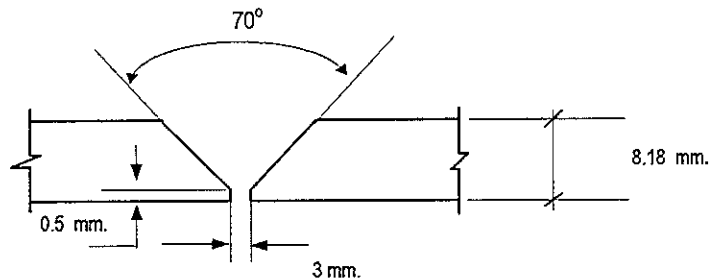
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : RNC-PQR-002

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	RNC-PQR-015	Date :	29 March 2013
WPS Number :	RNC-WPS-015		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW) + Shield Metal Arc Welding (SMAW)		
Type (Manual, Semi-auto, or Automatic) :	Manual		

JOINT DESIGN (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)				POSTWELD HEAT TREATMENT (QW-407)				
Material Spec.	A335	To	A335	Temperature	720° C			
Type or Grade	P11	To	P 11	Hold Time	2 HR			
P. No.	4	To	4	Others	Heating Rate 195° C/Hr Cooling Rate 250° C/Hr			
Thickness of Test Coupon		8.18 mm.		GAS (QW-408)				
Diameter of Test Coupon		Ø 8"						
Others								
FILLER METALS (QW-404)				Percent Composition				
					Gas (es)	Mixture	Flow Rate	
				Shielding	Argon	99.95%	10 L/Min.	
				Trailing	-	-	-	
				Backing	-	-	-	
				ELECTRICAL CHARACTERISTICS (QW-409)				
SFA Specification		GTAW	SMAW	Current	DC (GTAW)	DC (SMAW)		
AWS Classification		ER80S-G	E8016-B2	Polarity	EN	EP		
Filler Metal F-No.		6	4	Amperage	128 - 130 (GTAW)	Voltage	10 (GTAW)	
Weld Metal Analysis A-No.		3	3	102 - 120 (SMAW)		21 - 22 (SMAW)		
Size of Filler Metal		Ø 2.4 mm.	Ø 3.2 mm.	Tungsten Electrode Size		Ø 2.4 mm. 2% thoriated.EWTh-2		
Others		Kobelco TGS-1CM	Kobelco CMA-96	Others		N/A		
Weld Metal Thickness		-		TECHNIQUE (QW-410)				
POSITION (QW-405)				Travel Speed				5 - 10 cm/min.
Position of Groove		6G		String or Weave Bead		Both		
Progression of Welding (Uphill or Down)		Up Hill		Oscillation		N/A		
Others		N/A		Multiple or single Pass		Multiple Pass		
PREHEAT (QW-406)				Multiple or Single Electrodes		Multiple Electrodes		
Preheat Temperature		220° C		Others		N/A		
Interpass Temperature (max.)		260° C						
Others		N/A						

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number:- RNC-PQR-015

Tensile Test Report No.TS-13-04-029

Specimen	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (KN)	Ultimate Strength (N/mm ²)	Type of Failure & Loc.
1049/13 TR1	19.19	7.92	151.985	73.550	483.93	OUT OF WELD
1049/13 TR2	19.13	8.01	153.231	74.570	486.65	OUT OF WELD

Guided Bend Tests (QW-160) Report No.BD-13-04-012

Type and Figure Number	Type of Bend	Indication	Results
1049/13 BF1	Transverse face bend	No Open Discontinuity	Accepted
1049/13 BF2	Transverse face bend	Open Discontinuity 0.60,0.30,0.25 mm	Accepted
1049/13 BR1	Transverse root bend	No Open Discontinuity	Accepted
1049/13 BR2	Transverse root bend	No Open Discontinuity	Accepted

Toughness Tests (QW-170) Report No.IM-13-04-015

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Average (Joules)	Lateral Expansion Mils
1049/13-1	Weld Metal	V-Notch	- 20 °C	53.88	82.64	
1049/13-2	Weld Metal	V-Notch	- 20 °C	114.52		
1049/13-3	Weld Metal	V-Notch	- 20 °C	79.53		
1049/13-1	HAZ	V-Notch	- 20 °C	164.86	151.19	
1049/13-2	HAZ	V-Notch	- 20 °C	133.52		
1049/13-3	HAZ	V-Notch	- 20 °C	155.2		
1049/13-1	Base Metal	V-Notch	- 20 °C	122.17	137.6	
1049/13-2	Base Metal	V-Notch	- 20 °C	137.78		
1049/13-3	Base Metal	V-Notch	- 20 °C	152.86		

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A
Macroetch Test Results N/A

Other Tests

Type of Test Radiographic examination results Accepted Report No.PQR-002

Hardness Testing Report No. HV-13-04-006

Macrostructure Report No. MA-13-04-014

Welders Name Mr.Narongkon Y.

Welder Number W-013

Tests Conducted By Mr.Keerati Sa-ngoen

Laboratory Test LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Manufacture or Contractor :

Date :

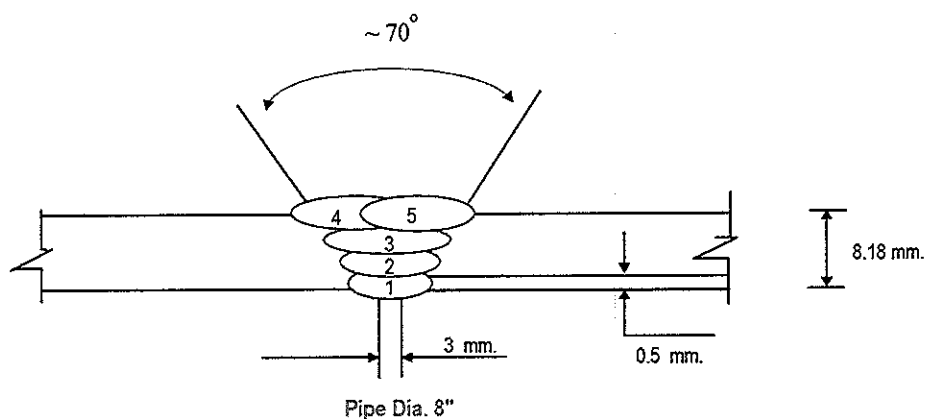
Inspected by : Mr.Wanlop N.

Certified by : Mr.Chaloemkiet R.

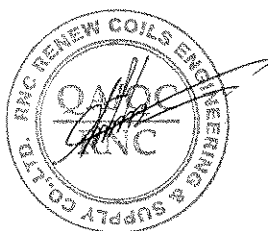
Date : 9 April 2013

PAE TECHNICAL SERVICE	WELDING PROCEDURE	PQR No. : RNC-PQR-015
		WPS No. : RNC-WPS-015
	QUALIFICATION RECORDS	THIS PQR RECORDS ACTUAL TEST
		CONDITION AND RESULT

Welding Sequences for RNC-PQR-015 (Butt Joint)



Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or Weave	Wire Feed	Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
							Bead	Speed				
1	GTAW	ER80S-G	2.4	128	10	DCEN	-	-	5	15.30	10	-
2	GTAW	ER80S-G	2.4	130	10	DCEN	-	-	5	15.60	10	-
3	SMAW	E8016-B2	3.2	102	21	DCEP	-	-	10	12.80	-	-
4	SMAW	E8016-B2	3.2	114	22	DCEP	-	-	10	15.00	-	-
5	SMAW	E8016-B2	3.2	120	21	DCEP	-	-	10	15.10	-	-





WELDERS CERTIFICATE



Record of Welder or Welding Operator Qualification Test

Client : PTT GLOBAL CHEMICAL CO., LTD. Project Name : PTTGC 2013
Contractor : Renewcoils Engineering & Supply Co., Ltd. Sub-Contractor : -

Welders Name : Mr. Narongkom Y. I.D. No. : 1 2512 00008 13 4 Welder No. : W-013
Welding Process (es) used : GTAW/SMAW Type : Manual
WPS Number (Used by welder during the welding of test coupon) : RNC-WPS-015
Base material (s) welded : ASTM A 335 P11 Thickness : 8.18 mm. (PQR)



(QW - 350)	Actual Values	Range Qualified
Backing (metal, weld metal, welder from both sides, flux, etc.) (QW-402)	With out Backing	With / With out Backing
ASME P - Number to ASME P- Number (QW - 403)	P-4 + P-4	P - No. 4
Base metal Dimension ☐ Plate ☒ Pipe (include diameter)	OD 8"	O.D 2-7/8" to Unlimited
Filler metal specification (SFA) A 5.28/5.5 Classification (QW - 404)	ER 80S-G/E8016-B2	F no.6+4
Filler metal F - Number	F6+4	F6+4
Consumable insert for GTAW or PAW	N/A	N/A
Weld deposit thickness for each welding process : GTAW /SMAW	3 / 5.18 mm.	Max. 6 / 10.36 mm
Welding position (QW - 405)	6G	All Position Unless Downhill
Welding Progression (uphill / downhill) (QW - 405)	Uphill	Uphill
Backing gas for GTAW , PAW , or GMAW ; Fuel gas for OFW (QW-408)	NO	YES or NO
GMAW transfer method / mode (QW - 409)	N / A	N / A
Welding current type and polarity	DC / EN	DC / EN

Visual Inspection Results (QW - 302.4)

Fit - up : Acceptable Root Pass : Acceptable Weld Profile : Acceptable

Guided Bend Test Results

Guided Bend Test Type	() QW - 462.2 (Side) Results	() QW - 462.3a (Trans. R&F Type)	() QW - 462.3b (Long. R&F) Results
-	-	-	-
-	-	-	-
-	-	-	-

Radiographic Test Results (QW - 304 & 305) ACCEPTABLE SEE REPORT No.PQR-002

(For alternative qualification of groove welds by radiography)

Fillet Weld - Fracture Test Result - Length and percent of defects - in.
Macro test fusion - Fillet leg size - in.x - in. Concavity / Convexity - in.
Welding test conducted by PAE TECHNICAL SERVICE CO.,LTD.
Mechanical test conducted by N / A Laboratory test No. N / A

We certify that the statements in this record are correct and that the test coupons were prepared . Welded

and tested in accordance with the requirement of : ASME - SECTION IX : 2010 / ASME B 31.3 : 2010

	Prepared By	Reviewed By / Approved By	Approved By
Signature :			
Name :	Mr. Chatchawan S.		
Date :	11/04/13		

PAE TECHNICAL SERVICE CO.,LTD.

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

TEL : 02) 322-0222 , FAX : 02) 721-2577 , RAYONG Tel : 038)682208-9, Fax : 038)682206, SRIRACHA : 038)330220, 330229

Page : 1 Of 1[illegible]



GLOBAL HEAT TREATMENT (THAILAND) CO.,LTD.

PREHEAT REPORT

CLIENT:	RENEW COILS ENGINEERING & SUPPLY CO.,LTD.		
PROJECT:	MINI SHUT DOWN PTTGC		
CONTRACTOR:	GLOBAL HEAT TREATMENT (THAILAND) CO.,LTD.		
PREHEAT METHOD:	☐ GAS	☒ ELECTRIC	COMPLETION DATE : 29-03-2013
LOCATION:	☒ SHOP	☐ FIELD	REPORT NO. PH -001
RECORDER NO:	EH- 07ZA194		PAGE NO. 2/2
CHART SPEED:	☒ 25 MM/Hr	☐ 50 MM/Hr	☐ 100 MM/Hr

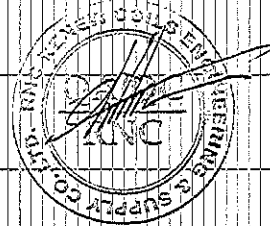
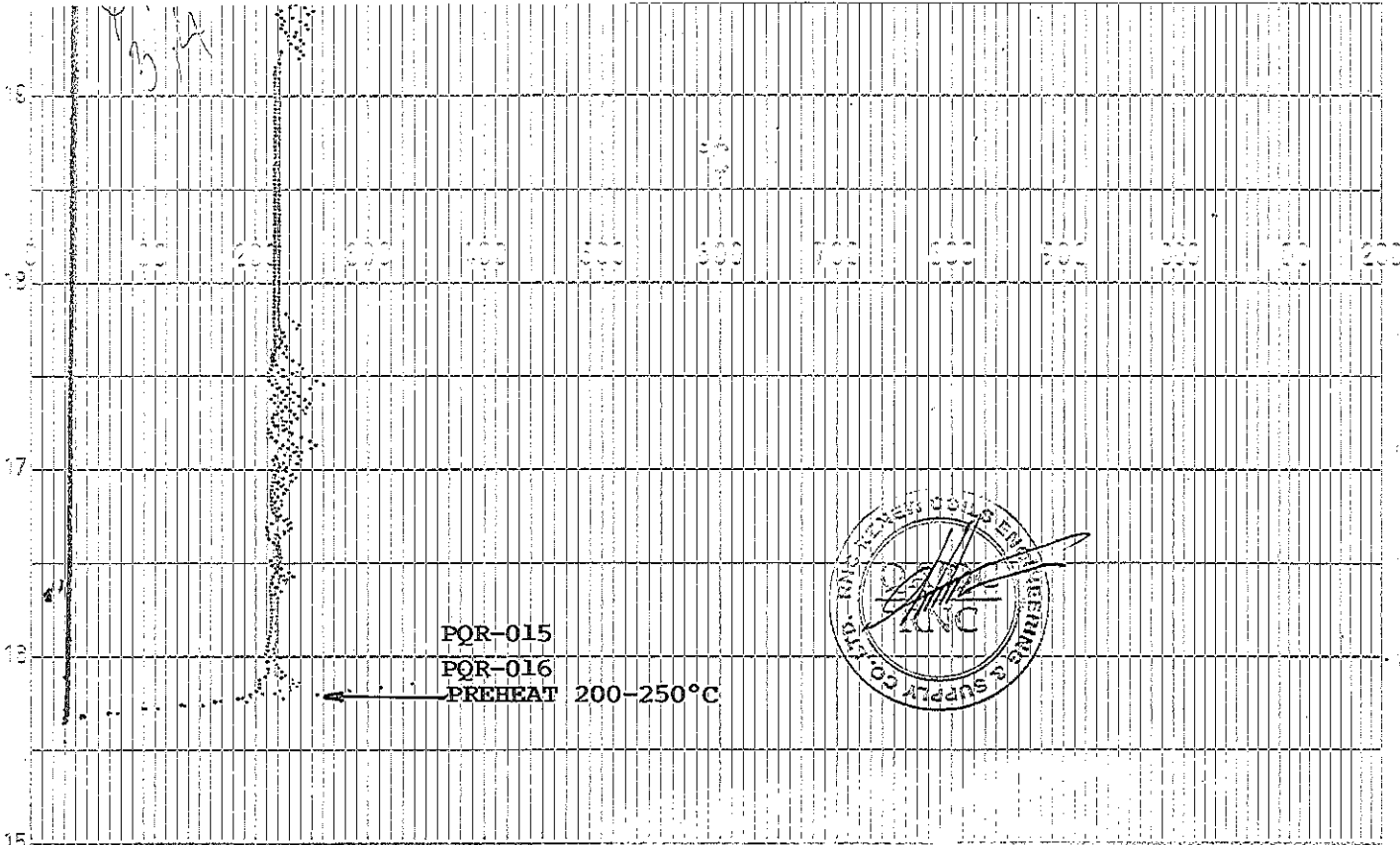
RESULT RECORDER CHART TO BE ATTACHED

DESCRIPTION	REQUEST	ACTUAL
1) PREHEAT TEMP (C° MIN)	200-250	220
2) INTERPASS TEMP (C° MAX)	300	260
3) POSTHEAT TEMP (C°)	-	-
4) POSTHEAT TIME (Hr)	-	-

MATERIAL : **A335 P11**

[illegible]

CHECKED BY GLOBAL HEAT TREATMENT (THAILAND)	APPROVED BY	APPROVED BY
SIGN: 	SIGN: 	SIGN: 
NAME: KOMKHUN D.	NAME: 	NAME: ภูชิน สีหอมไชย
DATE: 29-03-2013	DATE: 4/04/13	DATE: Phuchin Seehomchai



DATE. 29-03-2013

TC	COLOUR
1.	RED
2.	BLACK
3.	SKY BLUE
4.	GREEN

CHART NO. PH - 001

RECORDER NO. EH 07ZA194

CHART SPEED-25mm./HR.

CUSTOMER. RENEW COILS ENGINEERING & SUPPLY CO.,LTD.

PROJECT. MINI SHUT DOW PTGCG

PQR NO. RNC-PQR-014

SIZE. 10" THK. 9.27 mm.

PQR NO. RNC-PQR-015

SIZE. 8" THK. 8.18 mm.

PQR NO. RNC-PQR-016

SIZE. 8" THK. 8.18 mm.

MATERIAL. A335 P5, A335 P11

START TIME. 09.30 AM.

TECHNICIAN. MR. ANUSORN R.

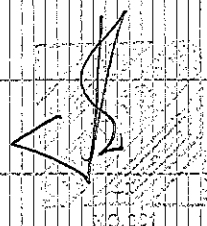


CHART NO. 21 201

NO. USMCHART

BOOK NO.

PAGE NO.



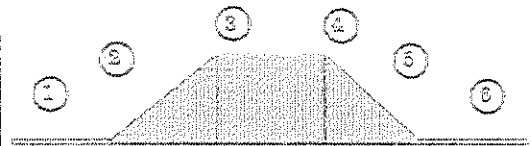
GLOBAL HEAT TREATMENT (THAILAND) CO., LTD.

HEAT TREATMENT REPORT

CLIENT:	RENEW COILS ENGINEERING & SUPPLY CO., LTD.		
PROJECT:	MINI SHUT DOWN PTIGC		
CONTRACTOR:	GLOBAL HEAT TREATMENT (THAILAND) CO., LTD.		
DESCRIPTION:	☒ PWHT ☐ NORMALIZE ☐ ANNEALING ☐ STABILIZE		
LOCATION:	☒ SHOP ☐ FIELD	REPORT NO.	PW-001
RECORDER NO:	EH- 0721194	PAGE NO.	1
CHART SPEED:	☒ 25 MM/Hr ☐ 50 MM/Hr ☐ 100 MM/Hr		
METHOD BY:	☒ ELECTRIC ☐ GAS ☐ OIL ☐ LOCAL ☐ FURNACE ☐ INTERNAL		
COMPLETION DATE:	30-03-2013	MATERIAL:	A335 P11

HEAT TREATMENT CONDITION / RESULTS

DISCRIPTION	REQUEST	ACTUAL
1) LOADING TEMP (C°)	300	300
2) HEATING RATE (C°/Hr)	220	195
3) HOLDING TEMP (MIN-MAX)	704-746	720
4) HOLDING TIME (Hr)	2 HR.	2 HR.
5) COOLING RATE (C°/Hr)	250	250
6) UNLOADING TEMP (C°)	300	300

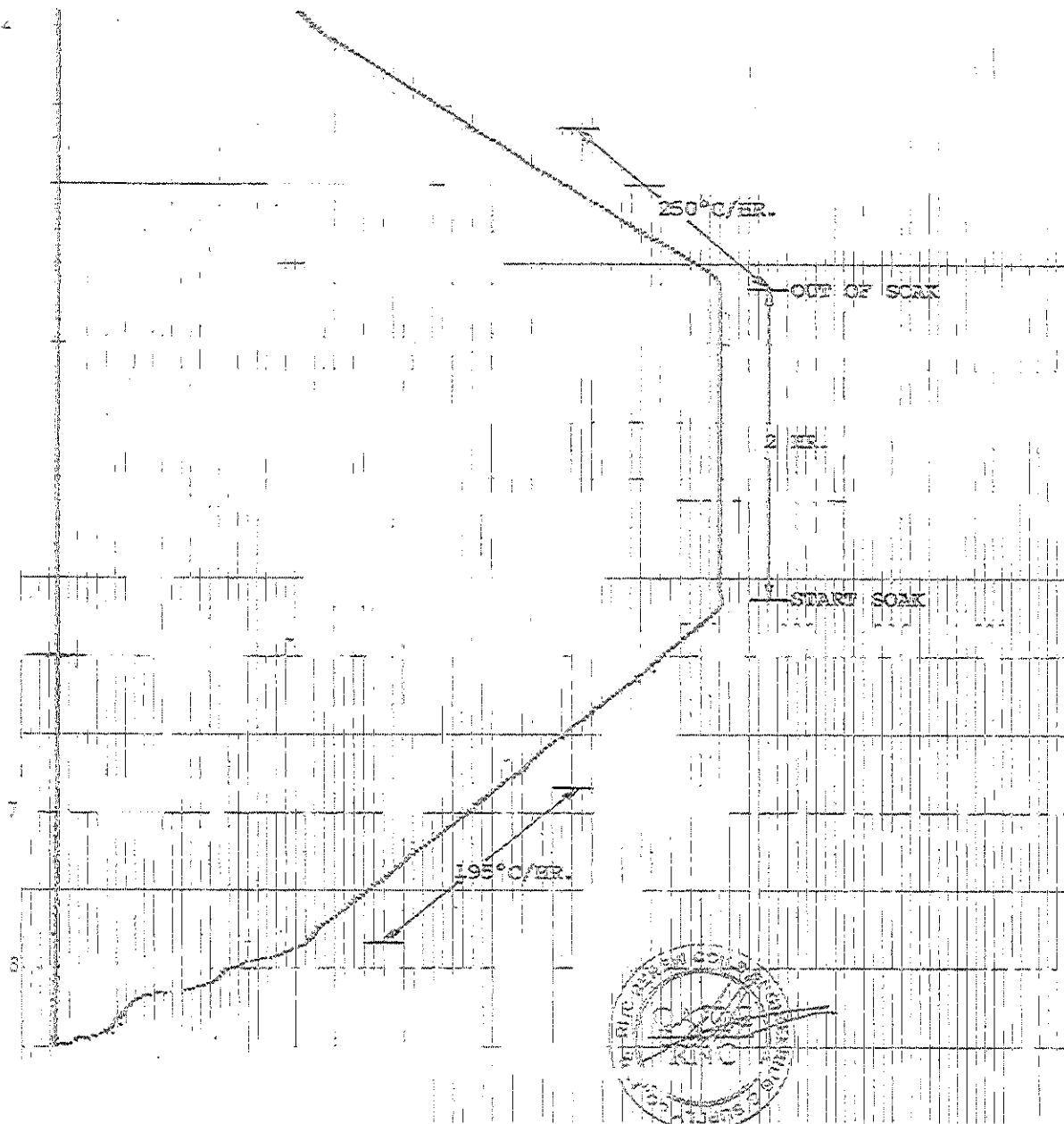


LINE NO. & CLASS & SHT NO. & SPL. NO. DWG. NO. & TANK NO.	WELD NO	SIZE	THK. (MM)	WELD TYPE	T/C NO.
PQR NO. RNC-PQR-015		8"	8.18	BW	1,2
PQR NO. RNC-PQR-016		8"	8.18	BW	3,4



TOTAL	2	JOINTS.
TOTAL	16	DS.

CHECKED BY GLOBAL HEAT TREATMENT (THAILAND)	APPROVED BY	APPROVED BY
SIGN:	SIGN:	SIGN:
NAME: KORKKOON D.	NAME: Phuchin Seehomchai	NAME: ภูชิน สีหอมไชย
DATE: 30-03-2013	DATE: 30/4/13	DATE: Phuchin Seehomchai



DATE. 30-03-2013

CHART NO. PW - 001

RECORDER NO. EH 0724194

CHART SPEED. 25mm./HR.

CUSTOMER. RENEW COILS ENGINEERING & SUPPLY CO., LTD.

PROJECT. MINI SHUT DOW PITGC

PQR NO. RNC-PQR-015

SIZE 8" THK. 8.18 mm.

PQR NO. RNC-PQR-016

SIZE 8" THK. 8.18 mm.

MATERIAL. A335/P11

START TIME. 10.00 AM.

TECHNICIAN. MR. CHAIKORN N.

NO.	COLOUR
1.	RED
2.	BLACK
3.	SKY BLUE
4.	GREEN

CHART NO. Z1 20

END



PAE TECHNICAL SERVICE CO.,LTD
NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT
 69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250
 Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 682208-9

WELDER PROCEDURE QUALIFICATION RECORD

Page No. 1 of 1

Client : Renew Coils Engineering & Supply Co.,Ltd.		Project Name : MINI SHUTDOWN PTTGC		Location : RNC Work Shop		Report No. PQR-002	
		Customer Order No. : N/A		WPS No. : RNC-WPS-015		Date : 29-Mar-13	

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
W-013	MR.NARONGKON Y.	GTAW / SMAW	6G	ASME IX	A335 P11	ø 8" Thk. 8.18 mm.	ER80S-G	ACCEPTED	ACCEPTED
							E8016-B2		

Joint Details

Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
1	GTAW	ER80S-G	2.4	DCEN	128	10	5	219	15.3	Argon	10	N/A	N/A
2	GTAW	ER80S-G	2.4	DCEN	130	10	5	220	15.6	Argon	10	N/A	N/A
3	SMAW	E8016-B2	3.2	DCEP	102	21	10	230	12.8	N/A	N/A	N/A	N/A
4	SMAW	E8016-B2	3.2	DCEP	114	22	10	241	15.0	N/A	N/A	N/A	N/A
5	SMAW	E8016-B2	3.2	DCEP	120	21	10	230	15.1	N/A	N/A	N/A	N/A

PAE Inspection By

SIGNATURE :
 NAME : Mr. Wanlop N.
 DATE : 30-Mar-2013

Witness By

SIGNATURE :
 NAME : Mr. Chatchawan
 DATE : 30-Mar-2013

Approved By

SIGNATURE :
 NAME : ภูชิน สอนมไชย
 DATE : Phuchin Seehomchai



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415 , E-mail : qa@lpnmp.co.th



8942

TESTING
No. 025

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. TS-13-04-029

Revision No. 0

Test Product	WPS No. RNC-WPS-015	Material Characterization	Pipe
Project Name	MINI SHUTDOWN PTTGC	Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A335 P11	Type of Test	Reduced Section Tension
Dimension	Ø 8" (8.18 mm Thk.)	Equipment's Capacity	200 TONS
Date Received	03 April 2013	Equipment / Serial No.	SHIMADZU UH200A / 32098926
Date Tested	05 April 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm ²)	Gauge Length (mm)	Load (kN)	Strength (N/mm ²)	Load (kN)	Strength (N/mm ²)		
1	1049/13 TR1	N/A	7.92	19.19	151.985	N/A	N/A	N/A	73.550	483.929	N/A	OUT OF WELD
2	1049/13 TR2	N/A	8.01	19.13	153.231	N/A	N/A	N/A	74.570	486.651	N/A	OUT OF WELD

The unit of N/mm² is equivalent to MPa

Additional details : Welding Process / Position : GTAW+SMAW / 6G

Welder Name : MR.NARONGKON Y.

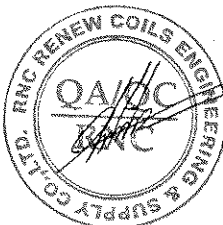
Welder No. : W-013

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date / /



Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date / /

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28945

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmp.co.th

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-13-04-012

Revision No. 0

Test Product	WPS No. RNC-WPS-015	Material Characterization	Pipe
Project Name	MINI SHUTDOWN PTGTC	Test Method	ASME Sec. IX : 2010 edition
Project No.	N/A	Standart Of Test	ASME Sec. IX : 2010 edition
Material Specification	A335 P11	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 8" (8.18 mm Thk.)	Diameter of Meudrel	32 mm
Date Received	03 April 2013	Bend Angle	180 Degree.
Date Tested	05 April 2013	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	1049/13 BF1	Transverse face bend	8.12x38.32x161.76 mm	No Open Discontinuity	N/A	-
2	1049/13 BF2	Transverse face bend	7.97x38.30x161.63 mm	Open Discontinuity	WELD ZONE	0.60, 0.30, 0.25 mm
3	1049/13 BR1	Transverse root bend	7.87x38.35x160.56 mm	No Open Discontinuity	N/A	-
4	1049/13 BR2	Transverse root bend	8.21x38.30x160.78 mm	No Open Discontinuity	N/A	-

Additional details : Welding Process / Position : GTAW+SMAW / 6G

Welder Name : MR.NARONGKON Y.

Welder No. : W-013

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date / /

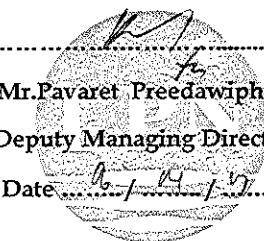


Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date / /

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8948

TESTING
NA 0131

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-13-04-015

Revision No. 0

Test Product	WPS No. RNC-WPS-015	Material Characterization	Pipe
Project Name	MINI SHUTDOWN PTTGC	Test Method	ASTM A370-10 and ASTM E23M-07a ^{cl}
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A335 P11	Type of Test	Charpy V-Notch Impact
Dimension	Ø 8" (8.18 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Date Received	03 April 2013	Equipment 's Capacity	542 Joules
Date Tested	05 April 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Specimen Dimension			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	1049/13-1	-20°C	5.002	10.007	54.87	53.88	82.64	Weld Metal	-
2	1049/13-2	-20°C	5.002	10.006	54.87	114.52			
3	1049/13-3	-20°C	5.000	10.009	54.86	79.53			
4	1049/13-1	-20°C	5.002	10.005	54.79	164.86	151.19	HAZ	-
5	1049/13-2	-20°C	5.003	10.006	54.83	133.52			
6	1049/13-3	-20°C	5.000	10.007	54.82	155.20			
7	1049/13-1	-20°C	4.999	10.007	54.86	122.17	137.60	Base Metal	-
8	1049/13-2	-20°C	5.001	10.009	54.89	137.78			
9	1049/13-3	-20°C	5.001	10.009	54.86	152.86			

Additional details : Welding Process / Position : GTAW+SMAW / 6G

Welder Name : MR.NARONGKON Y.

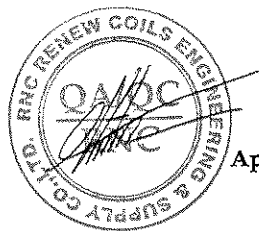
Welder No. : W-013

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 16 / 04 / 13



Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 16 / 04 / 13

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Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

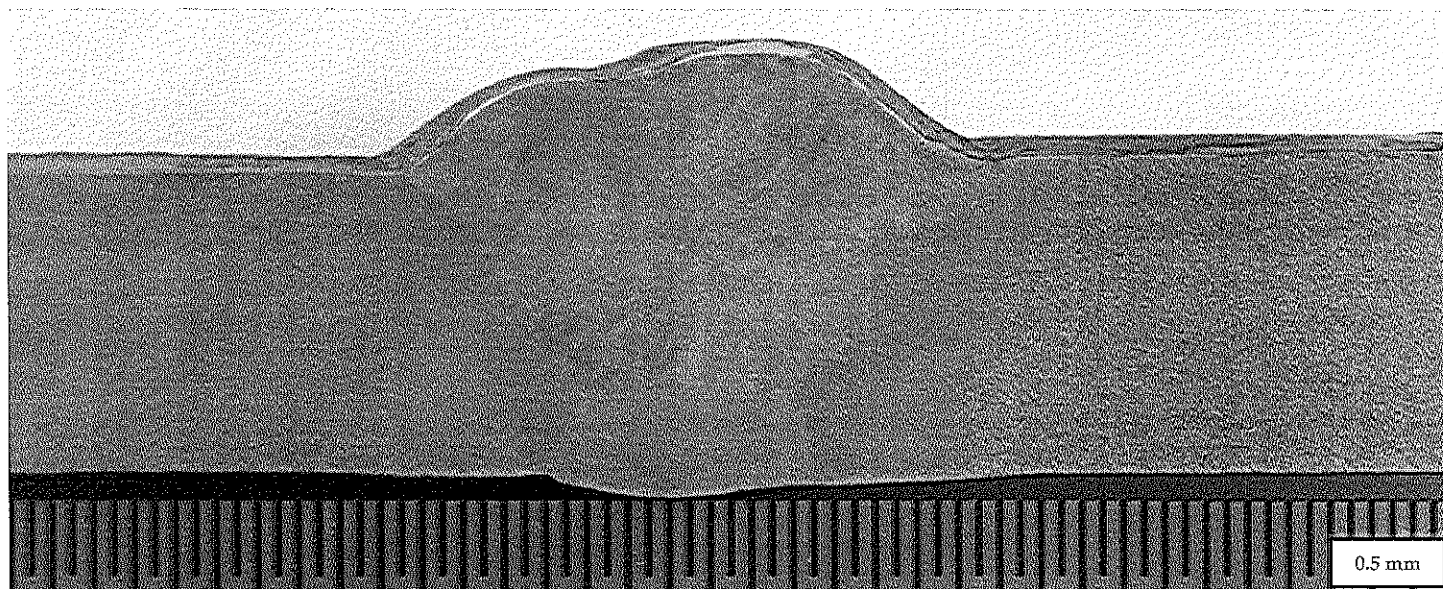
Macrostructure Analysis Report**Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.****Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.**

Page 1 of 1

Report No. MA-13-04-014

Revision No.0

Test Product	WPS No. RNC-WPS-015	Material Characterization	Pipe
Test No.	1049/13 MA	Equipment used	Microscope upto 40X (times)
Test Location	-	Serial No.	LEICA 0650854912
Material Specification	A335 P11	Method of Preparation	ASME Secction IX : 2010 edition
Dimension	Ø 8" (8.18 mm Thk.)	Test Method	ASME Section IX : 2010 edition
Welding Process/Position	GTAW+SMAW/6G	Standard of Test	ASME Secction IX : 2010 edition
Type of Joint	Butt Joint	Etchant Reagent	25 ml HNO ₃ + 75 ml H ₂ O
Welder Name/ No.	Mr. Narongkon Y. / W-013	Time of Etching	5 minutes
Date Received	3 April 2013	Date Tested	6 April 2013



Macro-etch Examination : The macrostructure shows complete fusion of the weldment. Weld metal and heat affected zone are free of cracks.

Additional details : Project Name : MINI SHUTDOWN PTTGC

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date : 09 / 04 / 13

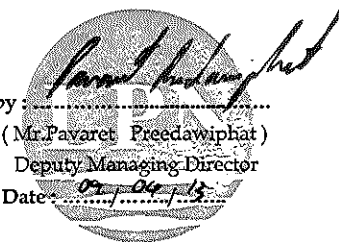


Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date : 09 / 04 / 13

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnrm.co.th

13292

Hardness Testing Report

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Page 1 of 1

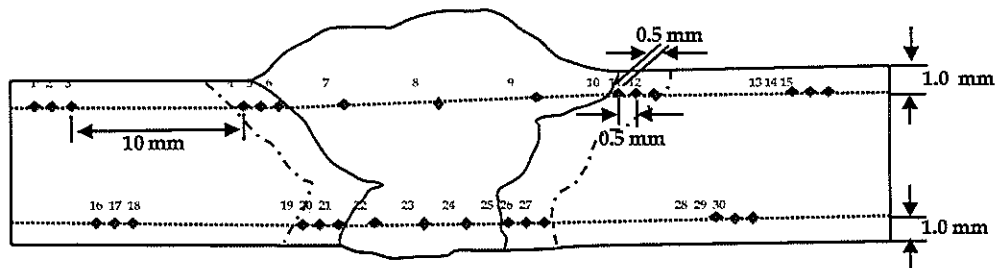
Report No. HV-13-04-006

Revision No. 0

Test Product	WPS No. RNC-WPS-015	Material Characterization	Pipe
Test No.	1049/13 HN	Test Method	ASTM E92-82 (Reapproved 2003) ^{e2}
Project Name	MINI SHUTDOWN PTTGC	Standard of	ASME Section IX : 2010 edition
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	A335 P11	Equipment	Wilson Wolpert , 432 SVD (V32D219)
Dimension	Ø 8" (8.18 mm Thk.)	Indenter Size	Pyramid diamond, angle of 136°
Welding Process/Position	GTAW+SMAW/6G	Test Load	HV 10 Kgf.
Type of Joint	Butt Joint	Total Force Dwell Time	15 s
Date Received	3 April 2013	Cal. Block	No. A60023 No. A62072
Date Tested	8 April 2013	Ambient Temperature	25.6 °C

Test Location

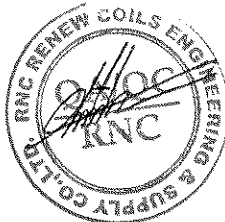
Total = 30 points



Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale)	Point No.	Location	Hardness value (Scale)
1	BASE	136.7	16	BASE	139.0	*****		
2	BASE	138.9	17	BASE	138.5			
3	BASE	138.1	18	BASE	140.0			
4	HAZ	144.4	19	HAZ	155.9			
5	HAZ	153.4	20	HAZ	160.4			
6	HAZ	168.3	21	HAZ	163.0			
7	WELD	208.7	22	WELD	183.3			
8	WELD	208.8	23	WELD	181.3			
9	WELD	207.6	24	WELD	168.5			
10	HAZ	190.9	25	HAZ	165.1			
11	HAZ	170.7	26	HAZ	165.0			
12	HAZ	151.7	27	HAZ	163.3			
13	BASE	140.9	28	BASE	140.0			
14	BASE	141.0	29	BASE	140.1			
15	BASE	139.1	30	BASE	140.0			

Additional details : Welder Name: Mr. Narongkon Y. / W-013

Reported by : *Paravit Preechawaphat*
 (Mr.Keerati Sa-ngoen)
 (Acting) General Manager - Quality Assurance
 Date : 09 / 04 / 13



Approved by : *Paravit Preechawaphat*
 (Mr.Pavaret Preechawaphat)
 Deputy Managing Director
 Date : 09 / 04 / 13

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SIGNATURE :	PAE INSPECTED BY	REVIEWED / WITNESSED BY	APPROVAL BY
			
	NAME : Mr. Anusorn M.	Mr. Chatchawan S.	ภูทิน สีหอมไชย Phuchin Seehomchai
DATE :	10-Apr-13	11/4/13	

ER 809-G

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INSPECTION CERTIFICATE CERTIFICATO DI COLLAUDO (UNI EN 10204 3.1 / ISO 10474 3.1 B)

Number / Numero: 11/04825
Page / Pagina: 1 / 8
Date / Data: March 11, 2011

Dati della S.p.A. - Sede legale e amministrativa:
Piazza Cavallotti 1844, 1
20124 Milano (MI) - Italy
Tel. 02 58 56 1111
Fax 02 58 56 3273

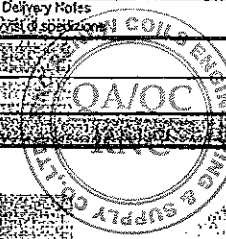
Customer / Cliente: Customer's Order Item / Ref. Ordine Cliente - Item VD00413080000273		Customer's Reference / Posizione Cliente: Manufacturer's Works Order N° / Conferma-Posizione 1210324/017	
Manufacture Process / Processo di fabbricazione: NORMALIZED AND TEMPERED NORMALIZZATI E RINVENUTI		Product Type / Tipo di prodotto: See note nr.1 Vedi nota nr.1	
Standard or Specification / Norme o specifiche: See note nr.3 Vedi nota nr.3		Steel Grade / Grado dell'Acciaio: STEEL GR. P 11 ASTM A 335 AND ASME SA 335 ACCIAIO GR. P 11 ASTM A 335 E ASME SA 335	
Dimensions / Dimensioni: d 219.1mm O.D. x 8.18mm W.T.		Length / Lunghezza: 10000 mm ± 11800 mm	
Schedule / Schede:		Quantity / Quantità: 50 Pcs/Pz. 191109° R 582.7 mt	
		Weight / Peso: 25241 kg	

DELIVERY NOTES / AVVISI DI SPEDIZIONE

Delivery Notes / Avvisi di spedizione:	Delivery Notes / Avvisi di spedizione:
Job number: 11503045/18	Job number: 10995 GRUGLIASCO TO
Shipping note: 00005559 - 11/03/2011	Country: ITALIA
Address: VIALE LIDICE, 30	

TENSILE TEST / PROVA DI TRAZIONE

Heat N° Cottura N°	Sample N° Prova N°	Lot N° Lotto N°	Specimen condition Condizioni della provetta				Specimen dimensions Dimensioni della provetta		Test Temp Temperatura di prova	Y.S.		U.T.S.		Elongation / Allungamento			
			ls	Se	Type	Dir	Size Dimensioni	Area Superficie		Min	Max	Min	Max	Ln	Ln	Min	Max
010659	R9613	1	B	NT	SS	L	38.00 x 7.90	301.90	20	386.00	386.00	517.00	517.00	2	2	29.5%	46.0%
010659	R9614C	1	B	NT	SS	L	38.00 x 7.90	301.90	20	338.00	338.00	518.00	518.00	2	2	29.5%	46.0%
010659	R9615C	1	B	NT	SS	L	38.10 x 7.90	302.50	20	369.00	369.00	495.00	495.00	2	2	29.6%	46.0%
010659	R9616C	1	B	NT	SS	L	38.10 x 8.00	306.40	20	340.00	340.00	497.00	497.00	2	2	30.0%	44.0%
010659	R9617	1	B	NT	SS	L	38.00 x 8.20	313.40	20	314.00	314.00	480.00	480.00	2	2	30.3%	46.0%
010659	R9618C	1	B	NT	SS	L	38.20 x 8.70	334.70	20	367.00	367.00	481.00	481.00	2	2	30.0%	43.5%
010659	R9619	1	B	NT	SS	L	38.00 x 8.40	321.00	20	324.00	324.00	509.00	509.00	2	2	30.7%	46.0%



Body / Corpo	Material / Materiale	Condition / Condizione	Test / Prova
Longitudinal / Longitudinale	Material / Materiale	Condition / Condizione	Test / Prova
Initial length / Lunghezza iniziale	Material / Materiale	Condition / Condizione	Test / Prova

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Your order n°: IP236915-18967
Our order n°: 03738012/PL It. 1A
Adv. of Dispm: 439280
Description: SPIRE PIPES DRL 8 STD P11

JORDI VAN BARLE
Heat: 910659
Qty: 116.5m

271322/JVD

Biz 1:8